

Chemical Engineering External Advisory Board Meeting Minutes

September 19, 2025, 9am-2pm

Engineering Innovation Building EIB 224

Attendees:

EAB Board: Saadet Acikgoz, Shailendra Bordawekar, Tarun Chandra, Claus Daniel, Soundar Kumaran, Dennis O'Brien,

Faculty: Vikas Berry, Alan Zdunek, Brian Chaplin, Said Al-Hallaj, Meenesh Singh

Meeting Notes (comments and suggestions):

General recommendations:

- Invite undergraduate and graduate students to Board meeting
- Invite the Dean and have more ChE faculty attend meetings.
- International exposure for engineering students is important- study abroad if possible. Note: There is an Engineering Abroad program from IES Abroad that was initiated last year but UIC engineering students have not yet participated.

State of the Department presentation (Berry):

- Universities that follow the National Academy of Engineering 14 Grand Challenges have shown increases in enrollment. Recommended that we add this component to our program.
- Get AIChE Chicago section more involved in our recruitment efforts
- Piggyback on AI/Computer Science trends in industry to increase our enrollment; Industry would rather hire chemical engineers who are fluent in AI/CS than a CS major who has to then learn ChE/engineering principles.
- Recommended adding the Argonne National Lab Center to the Entrepreneurship slide
- MS PharmE: Recommended contacting Eli Lilly to get input.
- Argonne uses pharmaceutical engineering process development principles in their R&D. There may be summer job opportunities at Argonne in this area.
- UIC Catalyst Center: Discussion on this new initiative. Board was favorable about this development. It was brought up that the agreements between industry and academia (UIC) need to be streamlined as there are challenges with UIC OTM rules and industry expectations. Also, there may be liability issues with students doing coursework projects at a plant if they are not employed by the company.
 - **Action items:** Need further industry input (plan a meeting) to bridge the gaps in industry/academic liaisons. Get more feedback from the advisers and discuss potential venues to enhance such industry collaborations and how we can facilitate further discussion with OTM and other stakeholders at UIC.

Undergraduate Program presentation (Zdunek):

ABET Review Update and Program Educational Objectives

- Overview of the ABET accreditation process that will be forthcoming in Fall 2026.
- Presented our Program Educational Objectives (PEOs)
 - **Action Item:** Send PEO's to Board for review and comment

Discussion on ChE concentrations and professional development

- Board generally agreed that the ChE concentrations are worthwhile and help brand/market our program. However, they are not necessarily the most important part when job searching.
- Board recommended these important skills/ experience to stand out in a job search:
 - Internships, undergraduate research
 - Excellent soft skills (selling yourself, communication both oral and written)
 - Examples of leadership in part-time jobs, extracurricular activities
 - Co-author on research publications
- Board recognized a gap in our program when professional development skills are taught - Freshmen year (ENGR100) and then senior year (CHE 499). Recommended adding professional development topics in sophomore and junior year courses.

Discussion on undergraduate curriculum improvement

- Adding CHE 101 Introduction to Chemical Engineering and Sustainability
Two proposals were discussed to open up 3 credit hours for CHE 101
 - Selective option between CME 260 Materials Properties (Materials Engineering course) and CHE 230 Molecular Systems in ChE
 - Selective option between CHEM 342 Physical Chemistry I and CHEM 222 Analytical Chemistry

Action item: Further discussion is needed. Zdunek will send more detailed information to the board on our proposal after meeting with the ChE Undergraduate Committee.
- Adding Data Science to our program
 - Discussed the merits of a BS ChE degree with a Data Science concentration or a Data Science major with a Minor in ChE.
 - Board recommended adding a dedicated course on data science to our BS ChE program. A data science concentration is not necessarily needed and the board agreed that a BS ChE major with data science skills is a better fit in the ChE industry than a data science or CS major with a minor in ChE.
- Recruitment discussion
 - Board recommended creating a UIC ChE Club that high school students can join and attend in order to foster interest in chemical engineering.

Graduate Program presentation (Chaplin):

Discussion on the merits of an MS ChE degree

- Discussed “What is student’s motivation for obtaining an MS ChE degree?”
 - Can’t find a job, an MS degree will help
 - Build up more skills to increase salary
 - Route to PhD
- Board commented that industry hires BS ChE’s or PhD ChE’s, MS ChE not as much. However, an MS ChE degree with an internship/co-op would be useful.
- Board recommended creating an AIChE Task Force amongst Chemical Engineering Heads to discuss trends and the success of MS ChE programs.

Graduate student enrollment

- In order to increase domestic student enrollment, board recommended building a network between the UIC ChE department and other universities

MS in Pharmaceutical Engineering Discussion

- Discussed target student groups for the MS PharmE degree:
 - Continuing education (offer night classes)
 - International students
 - Domestic non-ChE BS/MS majors: Pharmaceutical science, chemistry
 - Discussed the need to prepare non-ChE students by requiring pre-requisite courses and keeping separate requirements for BS ChE and non-ChE students.
 - Recommended offering night courses to attract more students.
- Discussion on “How value-added is this degree?”
 - Big pharma: Might be low as PhD’s are highly valued. MS PharmE students would be hired in ‘entry-level’ positions (this still could be good).
 - Biotech: Not sure, more research needed; ask industry for input
 - Recommended creating a survey to gauge interest in the degree through the AIChE Pharmaceutical Discovery, Development and Manufacturing Forum (PD2M)
 - Recommended to consider changing the name of the degree to broaden interest in the degree by supporting local (Chicago) strengths in biotechnology and pharmaceuticals.
 - Research value proposition and the target audience.
 - What programs at UIC are complimentary to the MS PharmE degree? (BME, PharmSci, Pharm). Recommended collaborating with these programs.